

INFO6007

Project Management in IT

Week 12: ITSM and processes 2

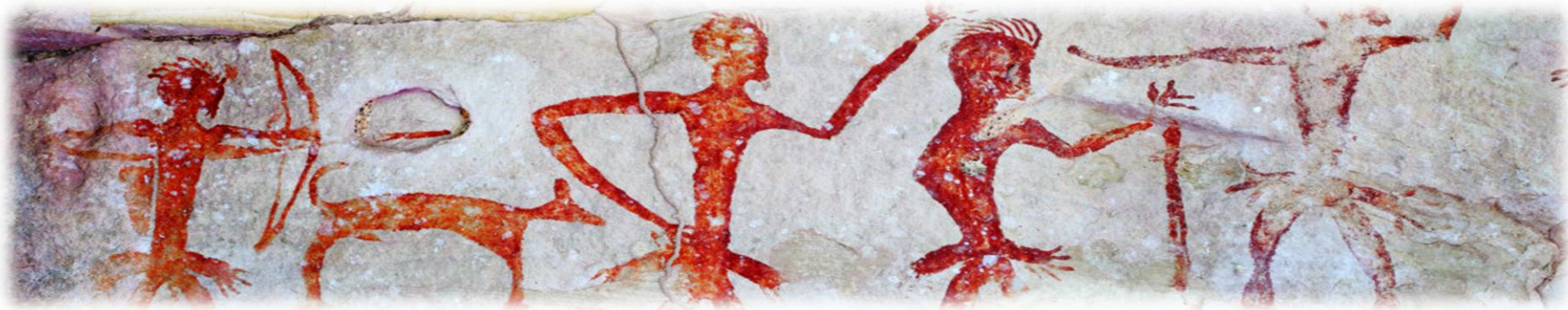
Semester 1, 2026



Acknowledgement of Country

I would like to acknowledge the Traditional Owners of Australia and recognise their continuing connection to land, water and culture. I am currently on the land of the Gadigal people of the Eora nation and pay my respects to their Elders, past, present and emerging.

I further acknowledge the Traditional Owners of the country on which you are on and pay respects to their Elders, past, present and future.



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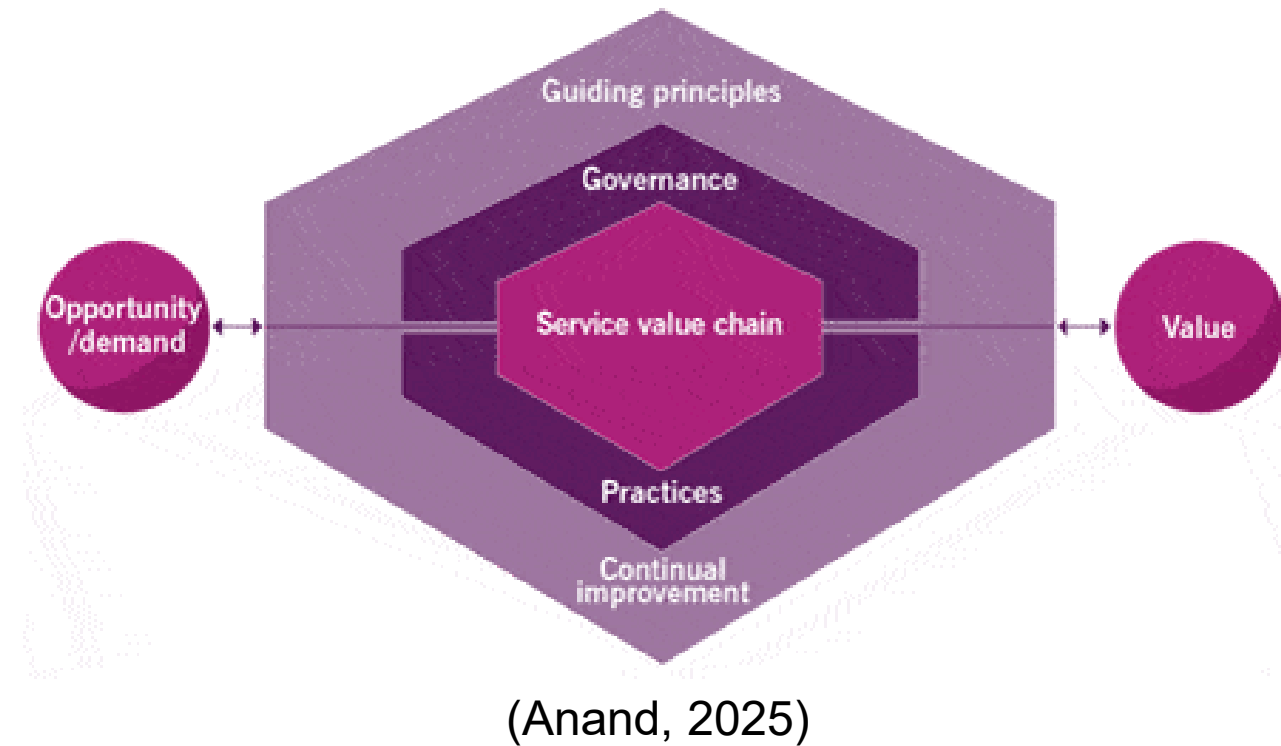
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Agenda

Recap

ITIL Overview

- **ITIL** stands for **Information Technology Infrastructure Library**, a **globally recognised framework** that provides **best practices for delivering high-quality IT services**.
- It defines **how IT services should be aligned with business needs**, ensuring consistency, efficiency, and continual improvement.



Key Components of ITIL v4.0

- Service Value System (SVS)
- Guiding Principles
- Governance
- Service Value Chain (SVC)
- ITIL 4 Practices
- Continual Improvement

Practices

- A **Practice** is a **set of organizational resources** (people, processes, information, tools, and partners) designed to perform specific work or achieve a specific objective.
- The ITIL 4 Practices provide **flexibility** and **adaptability**, allowing organizations to apply what's relevant depending on their context, rather than following rigid step-by-step processes.
- They support the **Service Value Chain** activities such as Plan, Design & Transition, Obtain/Build, Deliver & Support, and Improve.

Practices

- ITIL 4 defines **34 Management Practices**, grouped into **three categories**:
 - General Management Practices
 - Service Management Practices
 - Technical Management Practices
- Why practices matter?
 - Provide **flexibility** - adaptable across Agile, DevOps, or traditional environments.
 - Ensure **holistic improvement** by addressing people, processes, and technology.
 - Encourage **integration** rather than isolated process management.
 - Promote **continual improvement** across all organizational levels.

1. General Management Practices (14)

- Derived from business management disciplines, they support overall service management and governance.
- Examples:
 - Continual Improvement
 - Information Security Management
 - Risk Management
 - Knowledge Management
 - Project Management
 - Architecture Management
 - Relationship Management
 - Workforce and Talent Management

2. Service Management Practices (17)

- These are specific to IT service management and are key to value co-creation.
- Examples:
 - Incident Management
 - Problem Management
 - Change Enablement
 - Service Request Management
 - Service Level Management
 - Service Desk
 - IT Asset Management
 - Service Configuration Management
 - Release Management
 - Availability, Capacity, and Service Continuity Management

3. Technical Management Practices (3)

- Focus on the technical aspects of delivering IT services.
- Examples:
 - Deployment Management
 - Infrastructure and Platform Management
 - Software Development and Management

Key ITIL v4.0 Service Practices

- There are **17 Service Management Practices** in ITIL 4.
- These are the **core operational and strategic capabilities** that enable the creation, delivery, and continual improvement of IT services.
- The most important in these based on what we have learned so far are:
 - Incident Management
 - Problem Management
 - Change Management
 - Service Request Management
 - Service Desk
 - Service Level Management
 - Knowledge Management
 - Service Continuity Management

Key Service Practice - Incident Management

- **Incident Management** is the practice of **restoring normal service operation as quickly as possible** after an unplanned interruption, while minimizing the **impact on business operations**.
- Key activities:
 - **Incident Identification:** Detect or report incidents via Service Desk, monitoring tools, or users.
 - **Logging:** Record incident details in the ITSM tool for tracking and analysis.
 - **Categorization & Prioritization:** Assign type, severity, and urgency to focus on high-impact incidents first.
 - **Investigation & Diagnosis:** Determine the root cause or workaround.
 - **Resolution & Recovery:** Apply the solution or workaround to restore service.
 - **Closure:** Verify the service is restored and formally close the incident.
 - **Continual Improvement:** Analyze trends to prevent recurrence.

Key Service Practice - Problem Management

- **Problem Management** is the practice of **identifying, analyzing, and managing the root causes of incidents** to prevent recurrence and minimize the impact of future incidents.
- Key activities:
 - **Problem Detection:** Identify problems from recurring incidents, major incidents, or proactive trend analysis.
 - **Logging:** Document the problem with details, impact, and affected services.
 - **Categorization & Prioritization:** Assign urgency and impact to focus on critical problems first.
 - **Investigation & Diagnosis:** Determine root cause using tools like RCA (Root Cause Analysis).
 - **Workarounds:** Provide temporary solutions to reduce incident impact.
 - **Known Error Record:** Document the problem and workaround for reference.
 - **Resolution & Closure:** Implement permanent fix and verify effectiveness.
 - **Continual Improvement:** Review trends to prevent future problems.

Key Service Practice - Change Enablement

- **Change Enablement** is the practice of **ensuring that changes to services, systems, or infrastructure are assessed, approved, and implemented with minimal risk and disruption** to the organization.
- Key activities:
 - **Request for Change (RFC):** Capture all proposed changes formally.
 - **Assessment & Approval:** Evaluate risk, impact, resources, and alignment with business goals; approve or reject changes.
 - **Change Planning:** Schedule and coordinate implementation.
 - **Implementation:** Apply the change according to plan.
 - **Review & Close:** Verify success, document results, and update records.
 - **Emergency Changes:** Special fast-track process for urgent, high-priority changes.

Key Service Practice - Service Level Management

- **Service Level Management (SLM)** is the practice of **defining, agreeing, monitoring, and reporting on the performance of IT services** to ensure they meet the agreed business requirements and expectations.
- Key activities:
 - **Define SLAs:** Agree on measurable service targets (availability, response time, resolution time).
 - **Negotiate & Agree:** Align service commitments with business needs.
 - **Monitor & Report:** Track performance against SLAs and highlight breaches.
 - **Review & Improve:** Conduct regular service reviews and implement improvements.
 - **Coordinate with Other Practices:** Work with Incident, Problem, Change, and Service Desk practices to maintain agreed service levels.

Key Service Practice - Service Desk

- The **Service Desk** is the **single point of contact (SPOC)** between users and the IT organization, responsible for **handling incidents, service requests, and communication** with users.
- Key activities:
 - **Incident Logging:** Record incidents reported by users.
 - **Service Request Fulfillment:** Handle standard requests such as password resets or access provisioning.
 - **Incident Resolution / Escalation:** Resolve incidents or escalate to higher-level support teams.
 - **Communication:** Keep users informed about incident status and expected resolution times.
 - **Monitoring & Reporting:** Track tickets, SLAs, and performance metrics.
 - **Knowledge Management:** Update knowledge base articles to improve self-service.

Key Service Practice - Continual Improvement

- **Continual Improvement** is the practice of **aligning an organization's services, practices, and processes with changing business needs** by identifying and implementing improvements at all levels.
- Key activities:
 - **Identify Improvement Opportunities:** Gather data from incidents, problems, feedback, KPIs, and audits.
 - **Assess & Prioritize:** Evaluate potential improvements based on impact and value.
 - **Plan & Implement:** Develop and execute improvement initiatives.
 - **Measure & Review:** Track results, assess effectiveness, and make adjustments.
 - **Integrate Improvements:** Apply lessons learned to practices, processes, services, and value streams.

Service Value System

- The **Service Value System (SVS)** is the **core model in ITIL 4** that ensures **all components and activities of an organization work together to create value through IT-enabled services**.
- Guiding components of the SVS:
 - Guiding Principles
 - Universal recommendations that guide decisions and actions.
 - Governance
 - Ensures policies, authority, and accountability are in place. Provides direction and control over IT service delivery.
 - Service Value Chain
 - The operational core of the SVS. Represents **activities that transform demand into value**.
 - Practices
 - 34 management practices (General, Service, Technical) that provide capabilities to achieve outcomes.
 - Continual Improvement
 - Ensures **services, practices, and processes evolve** to meet changing business needs

Service Value Chain

- The **Service Value Chain** is a **set of interconnected activities** that an organization undertakes to **co-create value with customers**.
- The 6 key activities:
 - **Plan** – Align vision, strategy, and continual improvement initiatives.
 - **Improve** – Identify and implement improvements in services, practices, or processes.
 - **Engage** – Interact with stakeholders to understand demand, expectations, and feedback.
 - **Design & Transition** – Build, modify, and transition new or changed services into live use.
 - **Obtain/Build** – Acquire or build the components needed for service delivery.
 - **Deliver & Support** – Operate services and ensure ongoing support to meet expectations.

Value Streams

- A **Value Stream** is a series of steps an organization takes to create and deliver products or services that provide value to stakeholders.
- Key concepts:
 - Value Streams are **specific to outcomes** (e.g., onboarding a student, processing an LMS request).
 - They **integrate SVC activities and practices** to create efficient workflows.
 - Each step in the Value Stream should **contribute to value creation**, avoid waste, and optimize resources.
- Why does it matter?
 - Ensures **all parts of IT and the organization are aligned to value creation**.
 - Promotes **flexibility, transparency, and continual improvement**.
 - Bridges **strategy, operations, and customer outcomes**.

Common ITIL aligned tools – Service Now



- **ServiceNow** is a **cloud-based IT service management (ITSM) platform** that automates, manages, and integrates IT services across the enterprise.
- It is widely used to **implement ITIL-aligned practices** and streamline IT operations.
- Core capabilities:
 - **Incident Management:** Log, track, and resolve incidents efficiently.
 - **Problem Management:** Identify root causes and prevent recurring issues.
 - **Change Management / Change Enablement:** Plan, approve, and track changes with minimal risk.
 - **Service Request Management:** Automate user requests like password resets or software access.
 - **Reporting & Dashboards:** Monitor SLA compliance, incidents, and performance metrics.

Role of Automation and AI in ITSM

- Automation and Artificial Intelligence (AI) are transforming ITSM by **improving efficiency, reducing manual effort, and enabling faster, more accurate service delivery.**
- A few Key areas of impact are:
 - Incident management
 - Problem management
 - Change enablement
 - Service Request management
 - Monitoring and Event management

Role of Automation and AI in ITSM

Area of Impact	Impact
Incident Management	• Automated ticket routing to the correct support team. • AI-driven diagnosis to suggest probable causes or solutions. • Chatbots/self-service portals for instant resolution of common issues.
Problem Management	• AI analyzes incident trends to detect recurring problems. • Predictive analytics identifies potential risks before they impact services.
Change Enablement	• Automation manages routine changes with predefined workflows. • AI assesses risk, impact, and dependency analysis to support approvals.
Service Request Management	• Users submit requests via portals; automated workflows fulfill requests quickly. • Reduces manual approvals and repetitive tasks.
Monitoring and Event Management	• AI monitors system performance and triggers alerts for anomalies. • Predictive maintenance prevents outages before they occur.

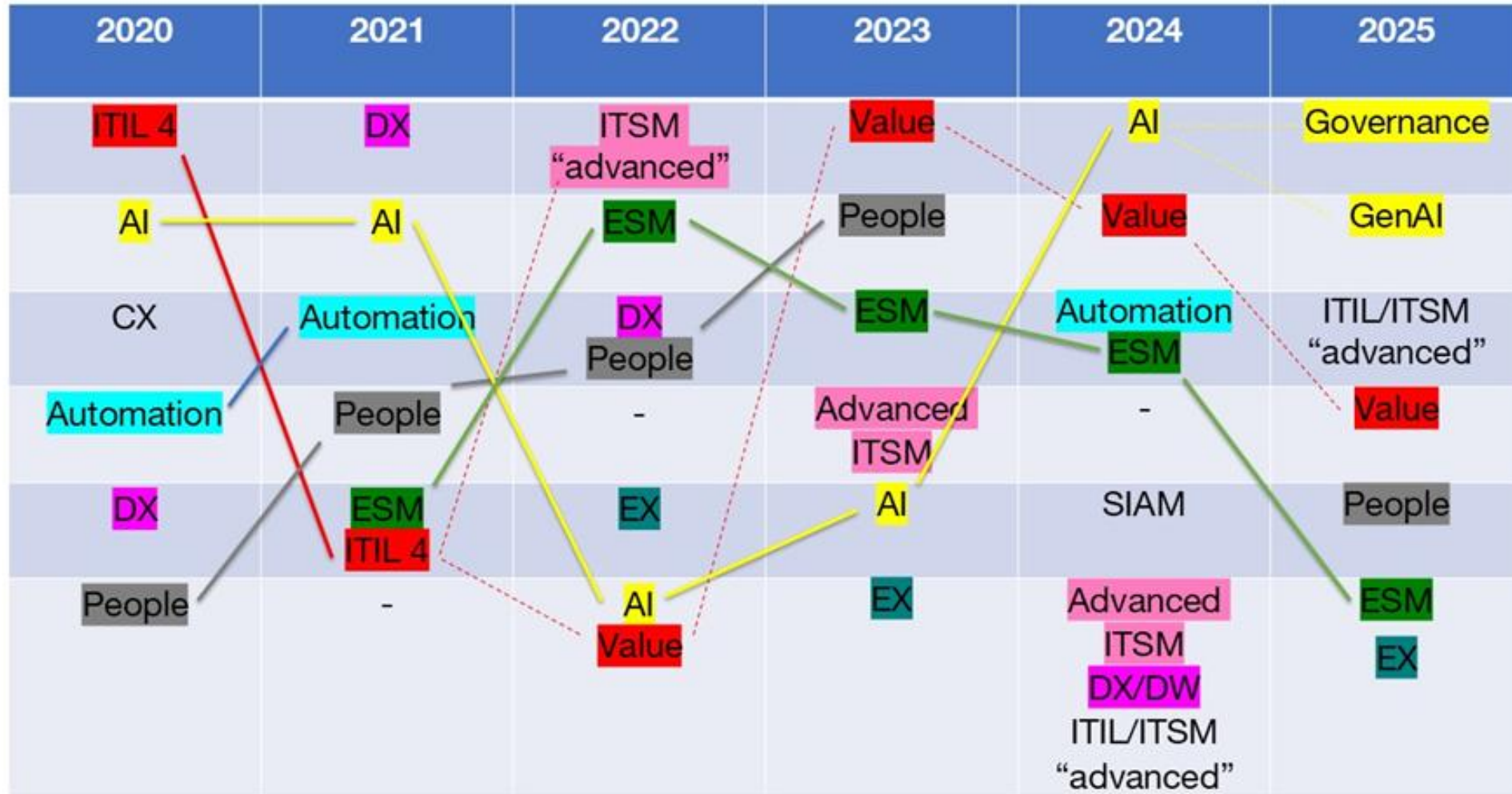
Challenges in ITIL adoption

- Cultural and Organisational resistance
 - Employees may resist **new processes, roles, and responsibilities**. ITIL adoption often requires a **shift from reactive to proactive service management**, which can face pushback.
- Complexity and Scope
 - ITIL 4 has **34 practices and multiple components** (SVS, SVC, guiding principles). Organizations may **overwhelm teams by trying to implement everything at once**, rather than focusing on value streams.
- Lack of Executive Support
 - Without **leadership commitment**, funding, and governance, ITIL initiatives struggle. Service improvement projects may stall or fail due to **insufficient prioritization**.

Current Trends

- Governance, Risk and Compliance (GRC)
 - Organizations are increasingly focusing on integrating GRC frameworks within ITSM tools to ensure compliance and manage risks effectively.
- Generative AI
 - The adoption of GenAI is revolutionizing ITSM by enhancing automation, improving decision-making, and providing personalized user experiences.
- Advanced ITIL/ITSM Capabilities
 - Organizations are moving beyond basic ITSM practices to implement advanced capabilities that offer greater flexibility and efficiency.
- Value Demonstration
 - There is a growing emphasis on demonstrating the value of ITSM initiatives to stakeholders, ensuring alignment with business objectives.

Current Trends



How does Project Management link to ITSM?

- While **IT Project Management (ITPM)** and **IT Service Management (ITSM)** have distinct objectives, they are **closely interconnected** in delivering IT value to organizations.
- **IT Project Management:** Focuses on **temporary initiatives** to deliver a specific output (e.g., new software, system upgrade).
- **ITSM:** Focuses on **ongoing delivery and support** of IT services to ensure continuous value creation.

How does Project Management link to ITSM?

- Project to Service transition
 - Projects often **create or modify IT services**. ITSM ensures these services are **operationalized, maintained, and supported** after project completion.
- Shared Processes and Practices
 - Both use **change management, configuration management, and knowledge management**. Projects need to **adhere to ITSM policies** to ensure smooth deployment and minimal disruption.
- Value continuity
 - Project success is measured by **delivering outputs**. ITSM success is measured by **delivering outcomes and value**. Linking the two ensures **outputs from projects generate lasting business value**.
- Feedback Loop
 - ITSM provides **performance data and lessons learned** back to projects for continual improvement. This ensures **future projects align better with operational needs**.

For You!

End of Lecture Questions

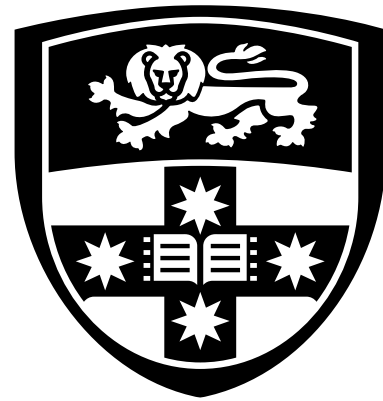
- Define **Incident Management** and explain its primary objective.
- How does Incident Management differ from Problem Management? Provide an example from a university IT environment.
- How can Problem Management reduce recurring incidents? Illustrate with an example.
- What is the purpose of **Change Enablement** in ITSM, and how does it minimize risk?
- How does Service Request Management improve user satisfaction and operational efficiency?
- Describe the **continual improvement model** in ITIL 4. How does it link to other practices?
- Explain how the **Service Value Chain** converts demand into value.

Case Study

- UniTech University is upgrading its **Learning Management System (LMS)** to include a new authentication module and online exam monitoring features. The IT department is responsible for ensuring **minimal disruption** to students and staff during the upgrade.
- During the rollout, several issues occur:
 - Students report login failures during peak hours.
 - The Service Desk is overwhelmed with incident tickets.
 - Some incidents repeat even after temporary fixes.
 - A critical change was deployed without full impact assessment, causing partial downtime.
 - IT management wants to **demonstrate value** and align the upgrade with SLA targets.

Case-study based questions

- Identify the incidents and problems in this scenario.
- How should the IT team handle recurring login failures using ITIL 4 practices?
- Discuss how proper change enablement could have prevented the partial downtime.
- What role should the Service Desk play during the LMS upgrade?
- Propose a **continual improvement plan** based on the lessons from this rollout.
- Map the **value stream** for resolving login issues from student report to service restoration.
- How can the IT team ensure project outputs create **long-term value** for students and staff?



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